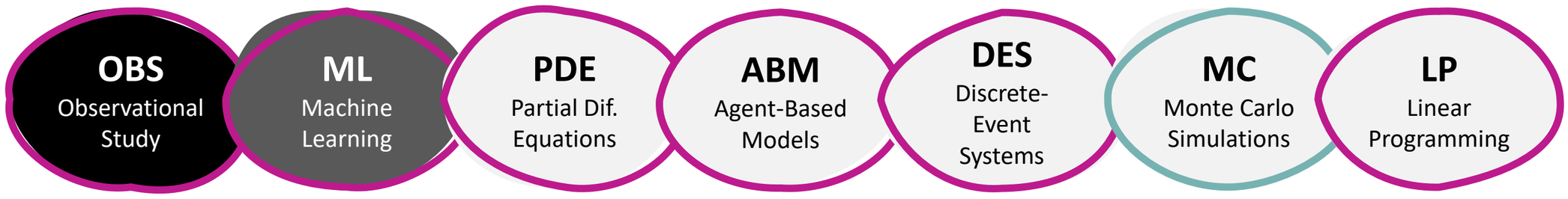
Most common modelling techniques



BEST FOR **This technique is specifically good at..**	Comparisons, patterns, what ius happening	Predict outputs from past examples	Smooth continuous change: growth, diffusion, spread etc	Studying emergent system behaviour from many interacting local entities	Queues, logistics, resources, workflows	Dealing with risks, uncertainties	Precise optimal solution under known constraints: resource allocation, scheduling etc
PURPOSE **Main goal of the modelling is ..**	Describe, explore, explain	Predict	Explore, explain		Explore, compute	Explore, estimate	Compute
TIME **How is the time dimension modelled**	No		Continous	Discrete (step- or event- based)		No (or included in model)	No
SPACE **How is the spatial dimension modelled**	No		Continous	Discrete	No or discrete	No (or in model)	
UNCERTAINTY **Degree of uncertainty in the system or goal?**	No	Implicit in data and pred errors	No	Yes, some randomness can be present	No / yes	Yes!	
MODELLING BASE **What do we need to know and specify about the system?**	Independent +dependent variables	Features, target, train/test data	Variables, initial conditions, dif equations	Agents + local rules, interaction rules, environment, initial population	Entities, resources, events, arrival patterns, queues, service times	Prob distributions of uncertain inputs, Model to run, # runs	Decision vars Constraints (equations) objective
RESULTS **Typical outcomes are ..**	Correlations, stats	Predictions, error metrics	Spatial patterns	Emergent patterns, clusters, metrics at system level	Waiting times, queue lengths, throughput, utilization	Outcome distributions	Optimal valuation
ANALYSIS **What is typically done with these models ?**	Visualization, Hypothesis testing	Train+test, predict, feature engineering	Simulation, sensitivity analysis	Scenario comparison, simulation	What-if, bottlenecks, policy comparison	Scenario comparison	Optimizations
TOOLS **Recommended tools and libraries**	Python (pandas, statsmodels) R ..	Scikit-learn	SciPy, P5.js, VisualPDE	P5.js, Mesa	SimPy, AnyLogic	Python (NumPy..)	Python (pulp)

Which ones are you using?